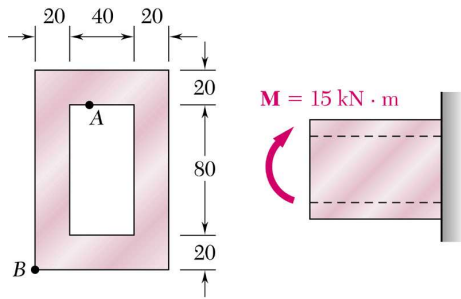


Zeshui Song, 3/21/26, HW5

4.1 Knowing that the couple  $M$  applied to the box beam as shown acts in a vertical plane, determine the stress at (a) point A, and (b) point B.



Dimensions in mm

Given:

- Neutral axis at geometric center (symmetric)
- Distance of A to neutral axis  $y_A = 40 \text{ mm}$
- Distance of B to neutral axis  $y_B = -60 \text{ mm}$
- $M = 15 \text{ kN}\cdot\text{m}$
- Moment of inertia for a rectangle about center  $I = \frac{1}{12}bh^3$
- $b_o = 80 \text{ mm}$ ,  $h_o = 120 \text{ mm}$
- $b_i = 40 \text{ mm}$ ,  $h_i = 80 \text{ mm}$

Find:

- Normal stress at A and B

Approach:

- Using superposition, I can find the moment of inertia of the box beam cross section (relative to neutral axis  $y=0$ )
- Find stress using the equation

$$\sigma = -\frac{My}{I}$$

1

Zeshui Song, 3/21/26, HW5

$$I_{outside} = \frac{1}{12}b_o h_o^3 = \frac{1}{12}[80 \text{ mm}][120 \text{ mm}]^3 = 11520000 \text{ mm}^4 = 0.00001152 \text{ m}^4$$

$$I_{inside} = \frac{1}{12}b_i h_i^3 = \frac{1}{12}[40 \text{ mm}][80 \text{ mm}]^3 = 1706666.66 \text{ mm}^4 = 0.000001707 \text{ m}^4$$

$$I = I_{outside} - I_{inside} = 9.813 \times 10^{-6} \text{ m}^4$$

(a) Normal stress at point A

$$\sigma_A = -\frac{My_A}{I} = -\frac{\left[15 \text{ kN} \cdot \text{m} \cdot \frac{10^3 \text{ N}}{1 \text{ kN}}\right] \left[40 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}}\right]}{[9.813 \times 10^{-6} \text{ m}^4]} = -61.1433 \text{ MPa}$$

(a) Normal stress at point B

$$\sigma_B = -\frac{My_B}{I} = -\frac{\left[15 \text{ kN} \cdot \text{m} \cdot \frac{10^3 \text{ N}}{1 \text{ kN}}\right] \left[-60 \text{ mm} \cdot \frac{10^{-3} \text{ m}}{1 \text{ mm}}\right]}{[9.813 \times 10^{-6} \text{ m}^4]} = 91.7150 \text{ MPa}$$

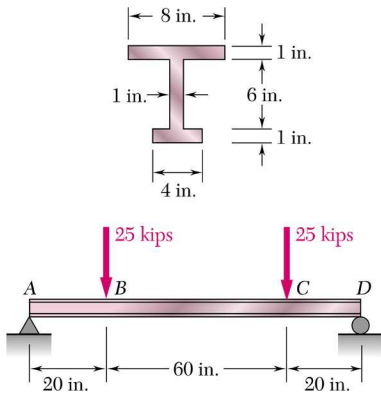
This makes sense because the beam in pure bending will bend with a center of curvature above the beam. Thus, point A will be under compression and B will be under tension. Also, since B is further from the neutral axis, it will see a larger stress. [Time spent: 30 mins]

2

Zeshui Song, 3/21/26, HW5

**Problem 4.8 will be graded**

**4.8** Two vertical forces are applied to a beam of the cross-section shown. Determine the maximum tensile and compressive stresses in portion BC of the beam.

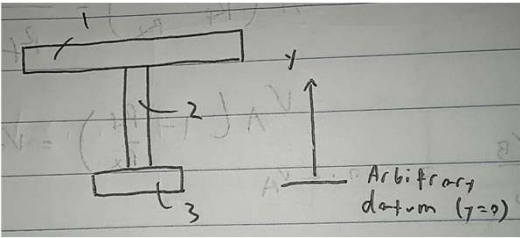


Given:

- $A_1 = 8 \text{ in}^2$ ,  $\bar{y}_1 = 7.5 \text{ in}$
- $A_2 = 6 \text{ in}^2$ ,  $\bar{y}_2 = 4 \text{ in}$
- $A_3 = 4 \text{ in}^2$ ,  $\bar{y}_3 = 0.5 \text{ in}$
- Force  $P = 25 \text{ kips}$

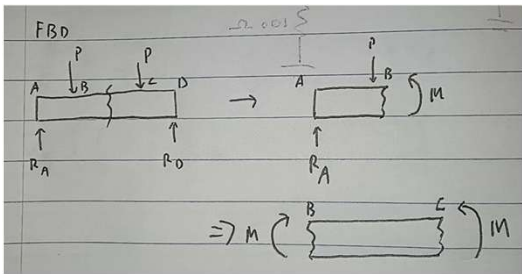
Find:

- Max tensile and compressive stress in BC



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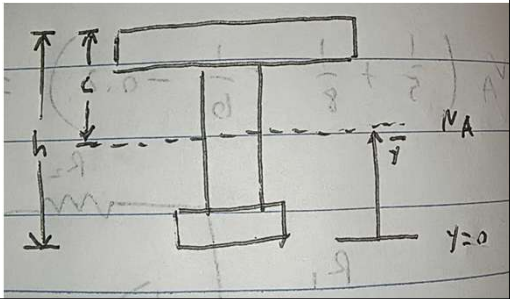
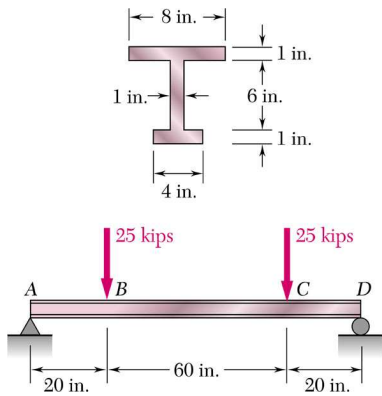
Approach:

- I can locate the neutral axis relative to the arbitrary datum using the equation

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i}$$

- Using superposition and parallel axis theorem, I can find the moment of inertia of the beam about the neutral axis
- I can use the FBD to determine the internal reaction moments due to the applied loads
- Finally, I can use the formulas for max tensile and compressive stress

$$\sigma_{m,c} = -\frac{Mc}{I}, \sigma_{m,t} = \frac{M(h-c)}{I}$$



4

Zeshui Song, 3/21/26, HW5

Find neutral axis

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2 + A_3 \bar{y}_3}{A_1 + A_2 + A_3} = \frac{[8 \text{ in}^2][7.5 \text{ in}] + [6 \text{ in}^2][4 \text{ in}] + [4 \text{ in}^2][0.5 \text{ in}]}{[8 + 6 + 4 \text{ in}^2]} = 4.777 \text{ in}$$

Find the total moment of inertia

$$I_{1,NA} = \frac{1}{12} b_1 h_1^3 + A_1 (\bar{y}_1 - \bar{y})^2 = \frac{1}{12} [8 \text{ in}][1 \text{ in}]^3 + [8 \text{ in}^2](7.5 - 4.777 \text{ in})^2 = 59.984 \text{ in}^4$$

$$I_{2,NA} = \frac{1}{12} b_2 h_2^3 + A_2 (\bar{y}_2 - \bar{y})^2 = \frac{1}{12} [1 \text{ in}][6 \text{ in}]^3 + [6 \text{ in}^2](4 - 4.777 \text{ in})^2 = 21.622 \text{ in}^4$$

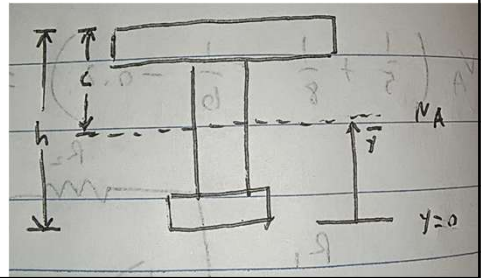
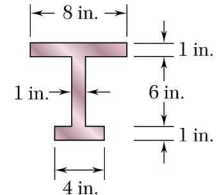
$$I_{3,NA} = \frac{1}{12} b_3 h_3^3 + A_3 (\bar{y}_3 - \bar{y})^2 = \frac{1}{12} [4 \text{ in}][1 \text{ in}]^3 + [4 \text{ in}^2](0.5 - 4.777 \text{ in})^2 = 73.264 \text{ in}^4$$

$$I = I_{1,NA} + I_{2,NA} + I_{3,NA} = 154.87 \text{ in}^4$$

Geometry

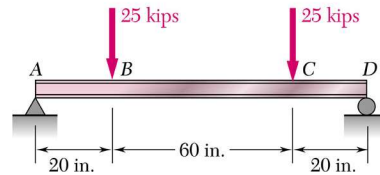
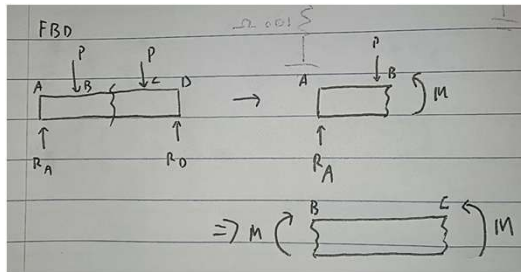
$$h = 1 + 6 + 1 \text{ in} = 8 \text{ in}$$

$$c = h - \bar{y} = 8 - 4.777 \text{ in} = 3.223 \text{ in}$$



5

Zeshui Song, 3/21/26, HW5



FBD analysis:

$$\sum M_A = -P(20 \text{ in}) - P(80 \text{ in}) + R_D(100 \text{ in}) = 0 \rightarrow R_D = P$$

$$\sum F_y = -2P + R_D + R_A = 0 \rightarrow R_A = P$$

Sum moments around B to find internal moment M

$$\sum M_B = -R_A(20 \text{ in}) + M = 0 \rightarrow M = P(20 \text{ in}) = \left[ 25 \text{ kips} \cdot \frac{10^3 \text{ lbf}}{1 \text{ kip}} \right] (20 \text{ in}) = 500000 \text{ lbf} \cdot \text{in}$$

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Zeshui Song, 3/21/26, HW5

Solve for stresses:

$$\sigma_{m,c} = -\frac{Mc}{I} = -\frac{[500000 \text{ lbf} \cdot \text{in}][3.223 \text{ in}]}{[154.87 \text{ in}^4]} = -10405.501 \text{ lbf/in}^2$$

$$\sigma_{m,t} = \frac{M(h-c)}{I} = \frac{[500000 \text{ lbf} \cdot \text{in}][8 - 3.223 \text{ in}]}{[154.87 \text{ in}^4]} = 15422.612 \text{ lbf/in}^2$$

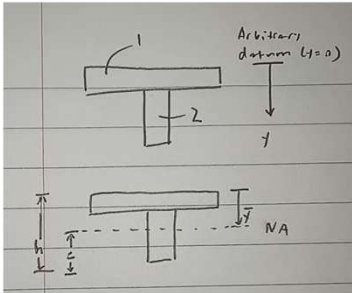
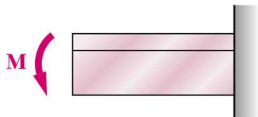
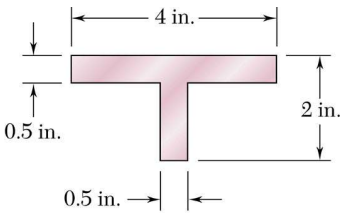
This makes sense because under the loading, section BC will be under pure bending with a center of curvature above the beam. Since there is less material in the region where the tensile stress is max, it makes sense that the max tensile stress will be greater than the max compressive stress. This could be an intentional design choice since the beam is not symmetrical. [Time spent: 40 mins]

7

Zeshui Song, 3/21/26, HW5

4.17 For the casting shown below, determine the largest couple M that can be applied without exceeding either of the following allowable stresses:  
 $\sigma_{all} = +6 \text{ ksi}$ ,  $\sigma_{all} = -15 \text{ ksi}$  (compression).

- Given:
- $A_1 = 2 \text{ in}^2$ ,  $\bar{y}_1 = 0.25 \text{ in}$
  - $A_2 = 0.75 \text{ in}^2$ ,  $\bar{y}_2 = 1.25 \text{ in}$
  - $\sigma_{all} = +6 \text{ ksi}$  (tension)
  - $\sigma_{all} = -15 \text{ ksi}$  (compression)
- Find:
- Max moment couple M



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Approach:

- I can locate the neutral axis relative to the arbitrary datum using the equation

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i}$$

- Using superposition and parallel axis theorem, I can find the moment of inertia of the beam about the neutral axis
- Finally, I can use the formulas for max tensile and compressive stress to find M from the allowable stresses

$$\sigma_{m,c} = -\frac{Mc}{I}, \sigma_{m,t} = \frac{M(h-c)}{I}$$

Find neutral axis

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2}{A_1 + A_2} = \frac{[2 \text{ in}^2][0.25 \text{ in}] + [0.75 \text{ in}^2][1.25 \text{ in}]}{[2 + 0.75 \text{ in}^2]} = 0.5227 \text{ in}$$

Find the total moment of inertia

$$I_{1,NA} = \frac{1}{12} b_1 h_1^3 + A_1 (\bar{y}_1 - \bar{y})^2 = \frac{1}{12} [4 \text{ in}][0.5 \text{ in}]^3 + [2 \text{ in}^2](0.25 - 0.5227 \text{ in})^2 = 0.1903 \text{ in}^4$$

$$I_{2,NA} = \frac{1}{12} b_2 h_2^3 + A_2 (\bar{y}_2 - \bar{y})^2 = \frac{1}{12} [0.5 \text{ in}][1.5 \text{ in}]^3 + [0.75 \text{ in}^2](1.25 - 0.5227 \text{ in})^2 = 0.5373 \text{ in}^4$$

$$I = I_{1,NA} + I_{2,NA} = 0.7276 \text{ in}^4$$

Geometry

$$h = 2 \text{ in}$$

$$c = h - \bar{y} = 2 - 0.5227 \text{ in} = 1.4773 \text{ in}$$

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Zeshui Song, 3/21/26, HW5

$$\sigma_{all,c} = \sigma_{m,c} = -\frac{Mc}{I}$$

$$M = -\frac{\sigma_{all,c} I}{c} = -\frac{\left[ -15 \text{ ksi} \cdot \frac{10^3 \text{ lbf/in}^2}{1 \text{ ksi}} \right] [0.7276 \text{ in}^4]}{[1.4773 \text{ in}]} = 7387.802 \text{ lbf/in}^2$$

$$\sigma_{all,t} = \sigma_{m,t} = \frac{M(h-c)}{I}$$

$$M = \frac{\sigma_{all,t} I}{(h-c)} = \frac{\left[ 6 \text{ ksi} \cdot \frac{10^3 \text{ lbf/in}^2}{1 \text{ ksi}} \right] [0.7276 \text{ in}^4]}{[2 - 1.4773 \text{ in}]} = 8352.0183 \text{ lbf/in}^2$$

Pick the smaller M

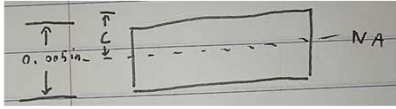
$$M = 7387.802 \text{ lbf/in}^2$$

This means that the region under compressive stress is the limiting region in this design. This makes sense because under this loading, the compressive region will be the underside of the beam which has less material at the max compressive stress region. [Time spent: 20 mins]

10

Zeshui Song, 3/21/26, HW5

**4.23** It is observed that a thin steel strip of 0.06-in. width can be bent into a circle of  $\frac{3}{4}$  in. diameter without any resulting permanent deformation. Knowing that  $E = 29 \times 10^6$  psi, determine (a) the maximum stress in the bent strip; (b) the magnitude of the couples required to bend the strip. (Discuss why this much bending doesn't result in permanent deformation and how this might be useful in design.)



Given:

- $b = 0.06 \text{ in}, h = 0.005 \text{ in}$
- $\rho = \frac{1}{2}D = \frac{1}{2}(0.75 \text{ in}) = 0.375 \text{ in}$
- $c = \frac{1}{2}h = \frac{1}{2}(0.005 \text{ in}) = 0.0025 \text{ in}$
- $E = 29 \times 10^6 \text{ psi}$

Find:

- Max stress in the bent strip
- Magnitude of moment couples  $M$

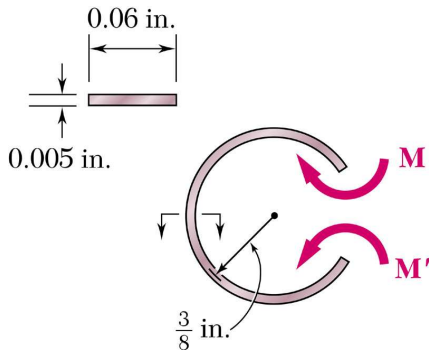
Approach:

- For the max stress, the magnitude of compressive and tensile stress is equal because it is symmetrical ( $h-c = c$ ) thus I can use the equation:

$$\sigma_{max,c} = -\frac{Ec}{\rho}$$

- For the moment couple, I can use the equation:

$$M = \frac{EI}{\rho}$$



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Moment of inertia:

$$I = \frac{1}{12}bh^3 = \frac{1}{12}[0.06 \text{ in}][0.005 \text{ in}]^3 = 6.25 \times 10^{-10} \text{ in}^4$$

(a) Max stress

$$|\sigma_{max}| = \frac{Ec}{\rho} = \frac{[29 \times 10^6 \text{ psi}][0.0025 \text{ in}]}{[0.375 \text{ in}]} = 193333.33 \text{ psi}$$

(b) Moment couple

$$M = \frac{EI}{\rho} = \frac{[29 \times 10^6 \text{ psi}][6.25 \times 10^{-10} \text{ in}^4]}{[0.375 \text{ in}]} = 0.04833 \text{ lbf} \cdot \text{in}$$

Because the strip is so thin, the max stress is very low since  $c = \frac{1}{2}h$  and max stress is proportional to  $c$ . Therefore, it can stay within the elastic region even while bending so much. This is also seen by the smaller moment couple required to bend it. This allows us to make flexible components without worrying about the part exceeding yield stress if it is thin enough. [Time spent: 15 mins]

12

Zeshui Song, 3/21/26, HW5

**2.68** A fabric used in air-inflated structures is subjected to a biaxial loading that results in normal stresses  $\sigma_x = 120 \text{ MPa}$  and  $\sigma_z = 160 \text{ MPa}$ . Knowing that the properties of the fabric can be approximated as  $E = 87 \text{ GPa}$  and  $\nu = 0.34$ , determine the change in length of (a) side AB, (b) side BC, (c) diagonal AC.

(Assume the fabric can be modeled as a homogeneous, isotropic, linearly elastic material.)

Approach:

- I can find the strain in x and z using the equations:

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y - \nu\sigma_z)$$

$$\varepsilon_z = \frac{1}{E}(-\nu\sigma_x - \nu\sigma_y + \sigma_z)$$

- I can find the change in lengths of the side lengths by

$$\delta_{AB} = (L_{AB})\varepsilon_x$$

$$\delta_{BC} = (L_{BC})\varepsilon_z$$

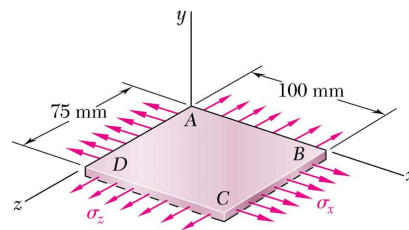
- The diagonal change in length can be found by geometry

Given:

- $\sigma_x = 120 \text{ MPa}$ ,  $\sigma_y = 0$ ,  $\sigma_z = 160 \text{ MPa}$
- $E = 87 \text{ GPa}$
- $\nu = 0.34$

Find:

- Change in length of sides AB, BC and the diagonal AC



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$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y - \nu\sigma_z) = \frac{1}{[87 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}}]} \left( \left[ 120 \text{ MPa} \cdot \frac{10^6 \text{ Pa}}{1 \text{ MPa}} \right] - [0.34] \left[ 160 \text{ MPa} \cdot \frac{10^6 \text{ Pa}}{1 \text{ MPa}} \right] \right) = 0.000754$$

$$\varepsilon_z = \frac{1}{E}(-\nu\sigma_x - \nu\sigma_y + \sigma_z) = \frac{1}{[87 \text{ GPa} \cdot \frac{10^9 \text{ Pa}}{1 \text{ GPa}}]} \left( -[0.34] \left[ 120 \text{ MPa} \cdot \frac{10^6 \text{ Pa}}{1 \text{ MPa}} \right] + \left[ 160 \text{ MPa} \cdot \frac{10^6 \text{ Pa}}{1 \text{ MPa}} \right] \right) = 0.00137$$

(a) Side AB

$$\delta_{AB} = (L_{AB})\varepsilon_x = [100 \text{ mm}][0.000754] = 0.0754 \text{ mm}$$

(b) Side BC

$$\delta_{BC} = (L_{BC})\varepsilon_z = [75 \text{ mm}][0.00137] = 0.1027 \text{ mm}$$

(c) Diagonal

$$L_{AC,i} = \sqrt{(L_{AB})^2 + (L_{BC})^2} = \sqrt{[100 \text{ mm}]^2 + [75 \text{ mm}]^2} = 125 \text{ mm}$$

$$L_{AC,f} = \sqrt{(L_{AB} + \delta_{AB})^2 + (L_{BC} + \delta_{BC})^2} = \sqrt{[100 + 0.0754 \text{ mm}]^2 + [75 + 0.1027 \text{ mm}]^2} = 125.1219 \text{ mm}$$

$$\delta_{AC} = L_{AC,f} - L_{AC,i} = 0.1219 \text{ mm}$$

This problem was mostly geometry, but it makes intuitive sense that the fabric will stretch and enlarge under tensile stress. [Time spent: 20 mins]

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